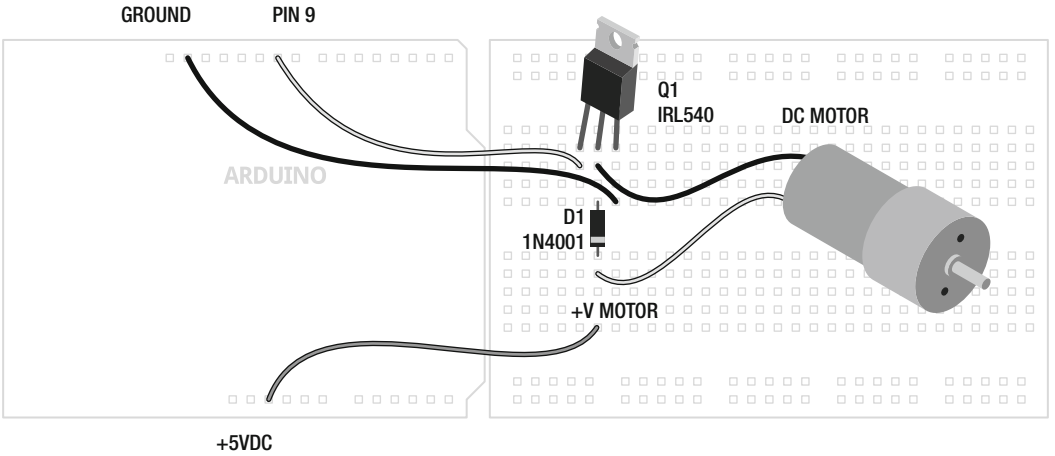


Figure 11-5. *MOSFET and DC Motor illustration*

When connected to pin 9 on the Arduino board, we can either control the motor digitally, meaning turn it on or off, using the `digitalWrite()` function or we could instead control the speed of the motor using PWM and the `analogWrite()` function with 0 being off and 255 being full speed forward. This high current and variable speed comes with the drawback that we can only control the motor in one direction.

H-Bridges

We have already used the SN754410 H-bridge integrated circuit in Chapter 9 to control a bipolar stepper motor, but we can also use it to control up to two DC motors, although we have only shown it with one in Figures 11-6 and 11-7. By using a couple of the Arduino's digital outputs, we can control the direction of the motor as well as its speed using one of the PWM pins. This, however, comes at the cost of the size of the motor, with the SN754410 only capable of 1-amp output capacity.